THE TENSOR PRODUCT OF SIMPLE ALGEBRAS, ONE OF THEM FINITE AND CENTRAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the reduction and the two steps that follow it each name the result they use, and no step is left to the reader.

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Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Wedderburn's theorem. Same chapter, Theorem theorem-wedderburn. Supplies the description of the finite central algebra as a matrix algebra over a finite skew field whose center is the ground field.
- Simplicity of a tensor product with a skew field. Same chapter, Lemma lemma-generate-two-sided-ideal. Supplies that tensoring a simple algebra with a skew field whose center is the ground field stays simple.
- Matrix algebras and simplicity. Same chapter, Lemma lemma-matrix-algebras. Supplies that a matrix algebra over a simple algebra is simple, which finishes the argument.

Source reference. Original theorem 0747: sources/stacks-brauer/pinned-original-source.tex, source lines 358–358. Navigation pointer invoked at proof line 519; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 074C: sources/stacks-brauer/pinned-original-source.tex, source lines 463–463. Navigation pointer invoked at proof line 549; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original lemma 074D: sources/stacks-brauer/pinned-original-source.tex, source lines 477–477. Navigation pointer invoked at proof line 522; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let A, A' be two simple k-algebras one of which is finite and central over k. Then $A \otimes_k A'$ is simple.*

Proof. Suppose that A' is finite and central over k. Write $A' = \mathrm{Mat}(n \times n, K')$, see Theorem 0747. Then the center of K' is k and we conclude that $A \otimes_k K'$ is simple by Lemma 074C. Hence $A \otimes_k A' = \mathrm{Mat}(n \times n, A \otimes_k K')$ is simple by Lemma 074D. $\square$